

Complex Analysis

Home Work 2

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Question 2.1 Look again at the subsets in Exercise 1.2(a) from Problem Sheet 1. Which of these sets are domains?

ANSWER:

Question 2.2 At which points $z \in \mathbb{C}$ are the following functions (a) continuous? (b) complex-differentiable?

(i) $\frac{1}{z^2 + 9}$

(ii) $z^{10} + 4\pi z^3 - 11$

(iii)

$$\begin{cases} \frac{z^2 - 1}{z^3 + 1}, & z \neq -1, \\ \frac{-2}{3}, & z = -1 \end{cases}$$

(iv) $(\operatorname{Re} z)(\operatorname{Im} z)$

ANSWER:

(i) Here we have a rational function - which is continuous at all points where the denominator is not zero.

The denominator is zero when: $z^2 + 9 = 0$, i.e. where $z = \pm 3i$. Hence, $\frac{1}{z^2+9}$ is continuous at all points of $z \in \mathbb{C} \setminus \{-3i, 3i\}$.

Being a rational function it is also differentiable for $z \in \mathbb{C} \setminus \{-3i, 3i\}$.

(ii) Here we have a standard polynomial function and hence this function is both continuous and differentiable for all $z \in \mathbb{C}$.

(iii)

Question 2.3

- (a) Show that the function

$$z \mapsto |z|^2$$

is complex-differentiable only at the point $z = 0$.

- (b) Let $U \subset \mathbb{C}$ be an open set. Suppose that $f : U \rightarrow \mathbb{C}$ is complex-differentiable at some point $z \in U$. Must f be holomorphic at z ?
- (c) At which points $z \in \mathbb{C}$ are the following functions (a) complex-differentiable, (b) holomorphic?

(i) $|z|$

(ii) $z|z|$

ANSWER:

- (a)

Question 2.4 Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be the function

$$f(z) = z^3.$$

Show that there is no point w on the line segment connecting 1 and i , i.e.

$$w \in \{ti + (1 - t)1 \mid t \in \mathbb{R}, 0 \leq t \leq 1\},$$

such that

$$\frac{f(i) - f(1)}{i - 1} = f'(w).$$

ANSWER:

Taking $f(z) = z^3$ we determine:

$$\begin{aligned} \frac{f(i) - f(1)}{i - 1} &= \frac{-i - 1}{i - 1} \\ &= \frac{1 + i}{1 - i} \\ &= \frac{1 + i}{1 - i} \frac{1 + i}{1 + i} \\ &= i \end{aligned} \tag{2.4.1}$$

Now, as $f'(z) = 3z^2$, we need to find a $w \in \{(1 - t) + ti \mid 0 \leq t \leq 1\}$ such that:

$$\frac{f(i) - f(1)}{i - 1} = 3w^2 \tag{2.4.2}$$

Using (2.4.1) and (2.4.2) and that they must be equal, we obtain:

$$3w^2 = i \implies 3[(1 - t) + ti]^2 = i$$

$$\implies 3[(1 - t) + ti]^2 = i$$

$$\implies 3\{(1 - t)^2 + 2t(t - 1)i - t^2\} = i$$

$$\implies (1 - 2t + t^2) + 2t(t - 1)i - t^2 = i.$$

$$\implies (1 - 2t) + 2t(t - 1)i = i.$$

real:

$$(1 - 2t) = 1 \Rightarrow t = 0.$$

IM:

$$2t(t - 1) = 1 \quad \text{no } t \in [0, 1] \text{ solves.}$$

$$\therefore \text{ no } w \in \{(1 - t) + ti \mid 0 \leq t \leq 1\} \text{ solves}$$

Question 2.5 Let Ω be a domain, and suppose that $z \in \Omega$. Show that

$$\Omega \setminus \{z\}$$

is a domain.

Now suppose that $r > 0$ is such that

$$D_r(z) \subseteq \Omega.$$

Need

$$\Omega \setminus D_r(z)$$

be a domain?

ANSWER:

Question 2.6 Given $c \in \mathbb{C}$, let the sequence $z_n(c)$ be defined iteratively in the following way:

$$z_0(c) = 0, \quad z_n(c) = z_{n-1}(c)^2 + c \quad \text{for } n \geq 1.$$

The *Mandelbrot set* M is the set of $c \in \mathbb{C}$ such that $|c| \leq 2$ and $|z_n(c)|$ does not tend to infinity as $n \rightarrow \infty$.

(i) Suppose that there exists $k \in \mathbb{N}$ such that

$$|z_k(c)| > 2.$$

Show that $c \notin M$.

(ii) Let U_k denote the set

$$U_k := \{c \in \mathbb{C} \mid |z_k(c)| > 2\}.$$

Show that U_k is open.

(iii) Deduce that M is a closed subset of $\mathbb{C}$.

ANSWER: